

Course Completion Summary

Professional Curriculum in Software QA, Automation, & Test Framework Design

September 2026 • 5 Courses Completed • 30 Total Hours

Curriculum Overview & Competencies

This summary records the completion of 5 intensive technical certifications spanning Python automation, browser control engines (Selenium & Playwright), Java enterprise test frameworks, and keyword-driven test architecture with Robot Framework.

Focus Area	Key Frameworks & Tools	Core Methodology
Web Automation	Selenium WebDriver, Playwright	Page Object Model (POM), Headless
Scripting & Testing	Python, Java, pytest, unittest	Data-Driven (CSV/Excel), CI/CD
Quality Assurance	Robot Framework, Tracing Analyzers	Root-Cause Diagnosis, Reporting

1. Python for Automation

- **Specialization:** Python Programming Fundamentals Specialization
- **Provider / Instructor:** Madecraft
- **Level:** Intermediate • **Effort:** 6 hours | 7 modules | 10 assignments
- **Timeline:** September 8, 2026 – September 11, 2026
- **Verification:** [View Certificate](#)

Course Overview – This course focuses on streamlining workflows by developing practical Python scripts for automated file handling, data parsing, and data validation. It covers web automation, API interaction, web scraping, and programmatically controlling web browsers.

Key Competencies & Tools Learned –

- **Core Skills:** Automation, File I/O & File Management, Web Scraping, Data Validation, Scripting, Authentication Handling, Unstructured Data Processing.
- **Tools & Technologies:** Python, Selenium, REST APIs, JSON, HTML, XML, CSS.

2. Introduction to Selenium

- **Specialization:** Software QA & Test Automation Engineering Specialization
- **Instructor:** Karlis Zars
- **Level:** Beginner • **Effort:** 3 hours | 1 module | 3 assignments
- **Timeline:** September 8, 2026 – September 11, 2026
- **Verification:** [View Certificate](#)

Course Overview – An introduction to browser automation and quality assurance engineering using Selenium. The curriculum emphasizes understanding core web automation concepts, creating automated test scripts, debugging workflows, and leveraging Selenium for both data-driven testing and web scraping tasks.

Key Competencies & Tools Learned –

- **Core Skills:** Test Automation, Test Script Development, Software QA & Testing, Debugging, Web Scraping, Test Data Management.
- **Tools & Technologies:** Selenium (Software), Cascading Style Sheets (CSS).

3. Selenium WebDriver with Python

- **Provider / Instructor:** Whizlabs / Whizlabs Instructor
- **Level:** Beginner • **Effort:** 8 hours | 3 modules | 10 assignments
- **Timeline:** September 8, 2026 – September 17, 2026
- **Verification:** [View Certificate](#)

Course Overview – A foundational course providing an in-depth understanding of Selenium WebDriver architecture and browser automation with Python. The curriculum covers setting up the development and testing environment, identifying and interacting with web elements, handling complex UI scenarios (such as popups, alerts, multiple browser tabs, and mouse/keyboard events), and structuring automated test suites using frameworks like unittest and pytest.

Key Competencies & Tools Learned –

- **Core Skills:** Test Automation, Unit Testing, Test Script Development, Software Testing & QA, Browser Compatibility, Debugging, Development Environment Setup, Software Architecture.
- **Tools & Technologies:** Python, Selenium WebDriver, pytest, unittest, HTML, CSS, Web Development Tools.

4. Selenium Automation and Testing Frameworks

- **Specialization:** Learn Automation Testing with Java and Selenium Webdriver Specialization
- **Provider / Instructor:** Packt - Course Instructors
- **Level:** Intermediate • **Effort:** 10 hours | 10 modules | 4 assignments
- **Timeline:** September 8, 2026 – September 22, 2026
- **Verification:** [View Certificate](#)

Course Overview – An in-depth course on building scalable test automation suites and industrial-grade testing frameworks using Java and Selenium. The curriculum focuses on cross-browser and headless execution strategies, implementing data-driven test architectures powered by CSV and Excel parsing, architecting modular test frameworks using the Page Object Model (POM), and distributing test execution across environments using Selenium Grid and standalone servers.

Key Competencies & Tools Learned –

- **Core Skills:** Test Automation, Software Testing, Test Script Development, Debugging, Data-Driven Testing, Maintainability, Test Data Management, Browser Compatibility, Test Architecture Design.
- **Tools & Technologies:** Java, Selenium (Software), Selenium Grid, Page Object Model (POM), Cascading Style Sheets (CSS), Data Import/Export (CSV/Excel).

5. Test Automation with Playwright (Python) & Robot Framework

- **Provider / Instructor:** Professionals in the Industry
- **Level:** Intermediate • **Effort:** 3 hours | 3 modules | 6 assignments
- **Timeline:** September 8, 2026 – September 23, 2026
- **Verification:** [View Certificate \(Coursera Verified\)](#)

Course Overview – This course provides comprehensive instruction on modern end-to-end browser automation and test framework design using Playwright and Robot Framework in Python. The curriculum focuses on architecting modular and maintainable test suites using the Page Object Model (POM), configuring test tagging and data parameterization for efficient CI/CD pipeline integration, diagnosing failures with traces, screenshots, and logs, and combining Playwright automation with readable Robot Framework keyword-driven testing.

Key Competencies & Tools Learned –

- **Core Skills:** Test Automation, Test Script Development, Quality Assurance (QA), Software Testing, Maintainability, Test Case Design, Debugging & Root-Cause Diagnosis, Continuous Integration / Continuous Deployment (CI/CD), Test Data Management, Test Execution Engine.
- **Tools & Technologies:** Python, Playwright, Robot Framework, CI/CD Pipelines, Trace & Execution Log Analyzers, Web Development Tools.